

Healthcare Insights: What Our Data Tells Us About Hospital Procedures and Patient Care

Engaging Opening Paragraph:

When I first started this project, I was amazed by how much data we have in healthcare. It seems like every visit to the hospital leaves behind a digital footprint. This realization reminded me of a recent visit to the emergency room with a family member. The hustle and bustle, combined with the healthcare professionals' dedication, got me thinking about the real stories hidden in hospital data. I wanted to uncover those stories and see what patterns could help improve patient care.

Why THIS Project?

I chose this project because I have a keen interest in healthcare analytics. I've seen how data can change lives, and I wanted to explore how it can influence medical practices. The goal was to analyze different variables in healthcare, which felt both relevant and impactful. It was exciting to see how analyzing such a large dataset might lead to insights that could benefit patients and medical professionals alike.

What Readers Will Gain:

In this article, readers will learn about the relationships between hospital stays, medical procedures, and medication usage. You'll see how this data can lead to improved understanding of patient needs and healthcare practices. The insights gained can help inform future healthcare decisions and policy-making.

Key Takeaways:

- The top five medical specialties are primarily related to cardiovascular health.
- More procedures lead to longer hospital stays.
- Race does not impact the number of procedures performed.
- Insulin is widely used across all age groups.

Dataset Details:

The dataset I used came from Kaggle and consists of 71,518 entries. It provides a comprehensive view of patient procedures, hospital stays, and medication usage. This wide range of data made it a suitable choice for analyzing healthcare trends and relationships.

Analysis Process:

I started by cleaning the data, ensuring that it was ready for analysis. Next, I looked at various relationships, such as the correlation between the number of procedures and time

spent at the hospital. I created visualizations to represent this data clearly. One surprising moment was finding that cardiovascular-related specialties dominated the list of top procedures. This led me to ponder why heart issues seem so prevalent among patients today.

Visuals and Insights:

- **Top Medical Specialties:** This visual highlights the five specialties with the most procedures. It's striking that so many focus on cardiovascular health, suggesting a significant patient need in this area.

```
select distinct(medical_specialty),  
round(avg(num_procedures),1) as  
avg_procedures, count(*) as  
count from patient.health  
group by medical_specialty  
having count > 50 and avg_procedures > 2.5  
order by avg_procedures DESC;
```

	medical_specialty	avg_procedures	count
►	Surgery-Thoracic	3.5	109
	Surgery-Cardiovascular/Thoracic	3.2	652
	Radiologist	3.2	1140
	Cardiology	2.7	5352
	Surgery-Vascular	2.6	533

- **Race and Procedures:** The lack of a relationship here indicates that race does not affect the number of procedures patients undergo, which is a promising finding for equitable healthcare.



```
select patient.demographics.race, avg(patient.health.num_lab_procedures) as avg_procedures from
patient.health
join patient.demographics on patient.health.patient_nbr=patient.demographics.patient_nbr
group by patient.demographics.race
order by avg_procedures;
```

	race	avg_procedures
▶	Asian	40.858934169278996
	Hispanic	42.71421495789995
	Caucasian	42.83813578251672
	Other	43.672572178477694
	?	43.996929824561406
	AfricanAmerican	44.07808105010939

- **Hospital Stay Predictions:** This graph illustrates how more procedures correlate with longer stays, useful for hospitals when estimating resource needs.



```
SELECT
AVG(time_in_hospital) AS avg_time
, CASE WHEN num_lab_procedures >=
AND num_lab_procedures < 25
THEN "few"
WHEN num_lab_procedures >= 25
AND num_lab_procedures < 55
THEN "average"
ELSE "many"
END AS procedure_frequency
FROM patient.health
GROUP BY procedure_frequency
ORDER BY avg_time DESC;
```

	avg_time	procedure_frequency
▶	5.657308841801269	many
	4.065429992110237	average
	3.27822257806245	few

- **Medication Rankings:** This chart shows that insulin ranks highest among all age groups, likely due to the prevalence of diabetes, which is an important health trend to recognize.

```
WITH long_health_cte AS (
SELECT patient_nbr, 'metformin' AS medication, metformin AS med_usage FROM patient.health
SELECT patient_nbr, 'glipizide' AS medication, glipizide AS med_usage FROM patient.health
SELECT patient_nbr, 'insulin' AS medication, insulin AS med_usage FROM patient.health
)
SELECT
age,
medication,
COUNT(*) AS total_uses,
RANK() OVER (PARTITION BY age ORDER BY COUNT(*) DESC) AS medication_rank
FROM long_health_cte
JOIN patient.demographics ON long_health_cte.patient_nbr = demographics.patient_nbr
WHERE med_usage NOT IN ('No')
GROUP BY medication, age
ORDER BY age, medication_rank;
```

	age	medication	total_uses	medication_rank
►	[0-10)	insulin	129	1
	[10-20)	insulin	487	1
	[10-20)	metformin	42	2
	[10-20)	glipizide	3	3
	[20-30)	insulin	934	1

Main Takeaways:

Through this analysis, I observed that cardiovascular health is a major concern, and more procedures correlate with longer hospital stays. The fact that race does not influence the number of procedures suggests a positive step toward equity in healthcare. The widespread use of insulin across all age groups indicates a significant public health issue. Understanding these patterns can help healthcare providers make informed decisions and improve patient outcomes.

Conclusion and Personal Reflections:

This project taught me the importance of data in understanding healthcare dynamics. One of the challenges I faced was ensuring the data's accuracy and relevance, which required careful attention. This experience has changed how I view data analytics in healthcare and has sparked my interest in continuing to explore this field.

Call To Action:

I'd love to hear your thoughts! Feel free to connect with me on LinkedIn. If you or someone you know is looking to hire a data analyst, I'm here to chat. Leave a comment with your questions or insights!